

Time Series Analysis
Practice Questions for TT1

Q. No	Question
1	State the differences between: AR, MA, ARMA, ARIMA and SARIMA model also write the general equation of each. • Definition • General equation • Role of parameters • When differencing is required • Example for each
2	Explain the concept of Unit Root in an AR(1) process: $X_t = \phi X_{t-1} + \varepsilon_t$ Discuss the behaviour of the series when: $\phi < 1$ $\phi = 1$ $\phi > 1$ Comment on stationarity in each case.
3	Write the general mathematical expressions for: (a) AR (2) model (b) MA(1) model
4	Explain the purpose of differencing in ARIMA modelling. When is first differencing sufficient?
5	Explain the SARIMA model. Interpret the parameters in SARIMA(p,d,q)(P,D,Q)_s.
6	A time series shows gradual exponential decay in ACF and sharp cutoff at lag 2 in PACF. Identify the model, write its equation, and justify theoretically.
7	ACF cuts off sharply after lag 1, while PACF shows gradual decay. Identify the model and explain the theoretical reason behind this behavior.
8	A monthly dataset shows significant spikes at lag 12 in ACF along with slow decay. Suggest an appropriate seasonal model and justify your answer.
9	An airline company observes: • Trend • Seasonality every 4 quarters • Autoregressive structure (a) Write general SARIMA equation (b) Explain parameters (p,d,q)(P,D,Q)_s (c) Suggest a suitable SARIMA model for quarterly airline data (d) Explain seasonal differencing.
10	Consider: $X_t = 15 + 0.5\varepsilon_{t-1} - 0.2\varepsilon_{t-2} + \varepsilon_t$ (a) Identify the model (b) Calculate: ρ_1, ρ_2 and ρ_3 (c) Sketch expected ACF and PACF
11	Explain the theoretical properties of an MA(2) model. Consider $X_t = 0.7\varepsilon_{t-1} + 0.3\varepsilon_{t-2} + \varepsilon_t$ and compute autocovariance at lag 1 and 2

12	Explain the role of ACF and PACF in model identification. How do they help distinguish AR, MA, and ARMA models using appropriate rules?																														
13	Explain the algorithm involved in time series forecasting from data initialization to model validation and diagnostic.																														
14	Define seasonal variation. Explain how seasonal differencing removes seasonality.																														
15	Explain the concept of autocorrelation. How is it interpreted using correlogram?																														
16	What is non-stationarity? Explain how unit roots cause non-stationarity.																														
17	Explain infinite past forecasting in time series. How is it different from Levinson–Durbin algorithm? (Not in TT1 but for practice)																														
18	(a) Explain forecasting using infinite past values. (b) Derive general expression for best linear predictor. (c) Explain the role of autocovariance matrix. (d) Compare with Levinson-Durbin Algorithm (Not in TT1 but for practice)																														
19	The annual production (in thousand units) of a factory is: <table><tr><td>Year</td><td>2016</td><td>2017</td><td>2018</td><td>2019</td><td>2020</td></tr><tr><td>Output</td><td>45</td><td>52</td><td>50</td><td>60</td><td>63</td></tr></table> 1. Fit a straight-line trend using Least Squares Method. 2. Estimate the trend values. 3. Forecast output for 2021. 4. Interpret the slope.	Year	2016	2017	2018	2019	2020	Output	45	52	50	60	63																		
Year	2016	2017	2018	2019	2020																										
Output	45	52	50	60	63																										
20	The quarterly sales (in ₹lakhs) of a retail company are: <table><tr><td>Year</td><td>Q1</td><td>Q2</td><td>Q3</td><td>Q4</td></tr><tr><td>2018</td><td>120</td><td>110</td><td>100</td><td>130</td></tr><tr><td>2019</td><td>125</td><td>115</td><td>105</td><td>135</td></tr><tr><td>2020</td><td>130</td><td>118</td><td>108</td><td>140</td></tr><tr><td>2021</td><td>135</td><td>120</td><td>112</td><td>145</td></tr></table> 1. Compute quarterly averages. 2. Compute the grand average. 3. Calculate seasonal indices 4. Identify peak seasonal quarter. 5. Comment on seasonal variation.	Year	Q1	Q2	Q3	Q4	2018	120	110	100	130	2019	125	115	105	135	2020	130	118	108	140	2021	135	120	112	145					
Year	Q1	Q2	Q3	Q4																											
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21	Calculate seasonal indices for the following data: <table><tr><td>Year</td><td>Q1</td><td>Q2</td><td>Q3</td><td>Q4</td></tr><tr><td>2012</td><td>85</td><td>78</td><td>74</td><td>88</td></tr><tr><td>2013</td><td>90</td><td>84</td><td>82</td><td>86</td></tr><tr><td>2014</td><td>88</td><td>80</td><td>76</td><td>92</td></tr><tr><td>2015</td><td>92</td><td>86</td><td>84</td><td>90</td></tr><tr><td>2016</td><td>94</td><td>88</td><td>86</td><td>96</td></tr></table> 1. Compute quarterly averages 2. Compute grand average 3. Calculate seasonal indices 4. Interpret highest seasonal quarter	Year	Q1	Q2	Q3	Q4	2012	85	78	74	88	2013	90	84	82	86	2014	88	80	76	92	2015	92	86	84	90	2016	94	88	86	96
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22	For the following parameter combinations, suggest appropriate model and justify: 1. (1,1,2) 2. (0,1,1)(1,1,1)₁₂ 3. (3,0) 4. (1,1,1)(0,1,1)₄ 5. (2,0,2)
23	Consider the AR(1) process: $X_t = 0.4X_{t-1} + \epsilon_t$ If the current observation is $X_t = 15$, 1. Find the one-step ahead forecast. 2. Find the two-step ahead forecast. 3. Find the three-step ahead forecast.
24	Explain the steps for using prediction in estimating time series.
25	Explain types of Time Series and its components.
26	Explain additive and multiplicative model with help of a graph and an example.
27	Difference between stationary and non-stationary.